

121. If the constant term in the expansion of $(1 + 2x - 3x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is p, then 108p is equal to _____.

यदि $(1 + 2x - 3x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ के प्रसार में अचर पद p है, तो 108p बराबर है _____

Ans. 54

Sol. General term of $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

$$T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(-\frac{1}{3x}\right)^r = {}^9C_r (-1)^r 3^{9-2r} 2^{r-9} x^{18-3r}$$

Constant term in expansion of $(1 + 2x - 3x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

$$= T_7 - 3T_8 = {}^9C_6 3^{-3} \cdot 2^{-3} + 3 {}^9C_7 \cdot 3^{-5} \cdot 2^{-2}$$

$$= \frac{3 \times 4 \times 7}{3^3 \cdot 2^3} + \frac{3 \times 9 \times 4}{3^5 \times 2^2} =$$

$$p = \frac{42 + 12}{108} = \frac{54}{108}$$

$$108p = 54$$

122. Remainder when $64^{32^{32}}$ is divided by 9 is equal to _____.

$64^{32^{32}}$ को 9 से विभाजित करने पर शेषफल है _____.

Ans. 1

Sol. Let $32^{32} = t$

$$64^{32^{32}} = 64^t = 8^{2t} = (9 - 1)^{2t}$$

$$= 9k + 1$$

Hence remainder = 1

123. The total number of six digit numbers, formed using the digits 4, 5, 9 only and divisible by 6, is _____.

केवल अंकों 4, 5, 9 के प्रयोग से बनायी गई छः अंको की संख्याएँ, जो 6 से विभाज्य है, की कुल संख्या है।

Ans. 81

Sol. Unit digit must be 4 since number should be divisible by 2.

Four out of remaining five places, each has 3 options and remaining one place will have only one option
so total number of six digit numbers = $3.3.3.3.1 = 81$

124. The coefficient of x^{2012} in the expansion of $(1-x)^{2008}(1+x+x^2)^{2007}$ is equal to _____.

$(1-x)^{2008}(1+x+x^2)^{2007}$ के प्रसार में x^{2012} का गुणांक बराबर है _____

Ans. 0

Sol. $((1-x)^{2008})[1+x+x^2]^{2007}$

$$(1-x)[1-x^3]^{2007}$$

$$(1-x)^{2007} C_r (-x^3)^r$$

$$^{2007}C_r [(-x)^{3r} + (-x)^{3r+1}]$$

$$\therefore 3r = 2012$$

$$\text{or } 3r + 1 = 2012$$

no such r exists

$\therefore$ Coefficient is zero

125. If the second, third and fourth terms in the expansion of $(x+y)^n$ are 135, 30 and $\frac{10}{3}$, respectively, then

$6(n^3 + x^2 + y)$ is equal to _____.

यदि $(x+y)^n$ के प्रसार में द्वितीय, तृतीय व चतुर्थ पद क्रमशः 135, 30 तथा $\frac{10}{3}$ हैं, तो $6(n^3 + x^2 + y)$ बराबर है _____

Ans. 806

Sol. ${}^nC_1 x^{n-1} y = 135$ _____(i)

${}^nC_2 x^{n-2} y^2 = 30$ _____(ii)

${}^nC_3 x^{n-3} y^3 = \frac{10}{3}$ _____(iii)

By $\frac{(i)}{(ii)}$

$$\frac{{}^nC_1 x}{{}^nC_2 y} = \frac{9}{2} \quad \text{_____ (iv)}$$

$$\text{By } \frac{\text{(ii)}}{\text{(iii)}}$$

$$\frac{{}^nC_2 x}{{}^nC_3 y} = 9 \quad \text{_____ (v)}$$

$$\text{By } \frac{\text{(iv)}}{\text{(v)}}$$

$$\frac{{}^nC_1 {}^nC_3}{{}^nC_2 {}^nC_2} = \frac{1}{2}$$

$$\frac{2n^2(n-1)(n-2)}{6} = \frac{n(n-1)}{2} \frac{n(n-1)}{2}$$

$$4n - 8 = 3n - 3$$

$$\Rightarrow n = 5$$

put in (v)

$$\frac{x}{y} = 9$$

$$x = 9y$$

put in (i)

$${}^5C_1 x^4 \left(\frac{x}{9} \right) = 135$$

$$x^5 = 27 \times 9$$

$$\Rightarrow x = 3, \quad y = \frac{1}{3}$$

$$6(n^3 + x^2 + y)$$

$$= 6 \left(125 + 9 + \frac{1}{3} \right)$$

$$= 806$$

126. Let $\alpha = \sum_{r=0}^n (4r^2 + 2r + 1) {}^nC_r$ and $\beta = \left(\sum_{r=0}^n \frac{{}^nC_r}{r+1} \right) + \frac{1}{n+1}$. If $140 < \frac{2\alpha}{\beta} < 281$, then the value of n is _____.

माना $\alpha = \sum_{r=0}^n (4r^2 + 2r + 1) {}^nC_r$ तथा $\beta = \left(\sum_{r=0}^n \frac{{}^nC_r}{r+1} \right) + \frac{1}{n+1}$ है। यदि $140 < \frac{2\alpha}{\beta} < 281$, तब n का मान होगा _____

Ans. 5

Sol. $\alpha = \sum_{r=0}^n r^2 {}^nC_r + 2 \sum_{r=0}^n r {}^nC_r + \sum_{r=0}^n {}^nC_r$

$$\sum (4 \cdot r(r-1) + 6r + 1) {}^nC_r$$

$$4 \cdot n(n-1) 2^{n-2} + 6 \cdot n \cdot 2^{n-1} + 2^n$$

$$= 2^n (n+1)^2$$

$$\beta = \sum_{r=0}^n \frac{{}^nC_r}{r+1} + \frac{1}{n+1} = \frac{1}{n+1} \cdot [2^{n+1} - 1] + \frac{1}{n+1} = \frac{2^{n+1}}{n+1}$$

$$\text{Now, } 140 < \frac{2\alpha}{\beta} < 248 \quad 140 < (n+1)^3 < 281$$

$$n+1 = 6 \quad n = 5$$

127. The sum of all rational terms in the expansion of $\left(1 + 2^{\frac{1}{3}} + 3^{\frac{1}{2}}\right)^6$ is equal to _____.

$$\left(1 + 2^{\frac{1}{3}} + 3^{\frac{1}{2}}\right)^6 \text{ के प्रसार में सभी परिमेय पदों का योग बराबर है } \underline{\hspace{2cm}}.$$

Ans. 612

Sol. $\left(1 + 2^{\frac{1}{3}} + 3^{\frac{1}{2}}\right)^6$

$$\frac{|6|}{|p|q|r|} (1)^p \times (2)^{\frac{q}{3}} 3^{\frac{r}{2}}$$

p	q	r
6	0	0
4	0	2
2	0	4
0	0	6
3	3	0
1	3	2
0	6	0

total seven term are rational

128. If $f(x)$ and $g(x)$ are two polynomials such that the polynomial $P(x) = f(x^3) + x g(x^3)$ is divisible by $x^2 + x + 1$, then $P(1)$ is equal to _____ .

यदि $f(x)$ तथा $g(x)$ दो बहुपद हैं जिनके लिए बहुपद $P(x) = f(x^3) + x g(x^3)$, $x^2 + x + 1$ से विभाज्य है, तो $P(1)$ बराबर होगा _____ ।

Ans. 0

Sol. $P(x) = f(x^3) + x g(x^3)$

$$P(1) = f(1) + g(1) \quad \dots\dots\dots(1)$$

$$P(x) = Q(x) (x^2 + x + 1)$$

$$P(\omega) = 0 = P(\omega^2)$$

ω and ω^2 are non - real

cube roots of unity

$$P(\omega) = f(\omega^3) + \omega g(\omega^3)$$

$$f(1) + \omega g(1) = 0 \quad \dots\dots\dots(2)$$

$$P(\omega^2) = f(\omega^6) + \omega^2 g(\omega^6) = 0$$

$$f(1) + \omega^2 g(1) = 0 \quad \dots\dots\dots(3)$$

$$(3) + (2)$$

$$\Rightarrow 2f(1) + (\omega + \omega^2)g(1) = 0$$

$$2f(1) = g(1) \quad \dots\dots\dots(4)$$

$$(2) - (3)$$

$$\Rightarrow (\omega - \omega^2) g(1) = 0$$

$$g(1) = 0 = f(1) \text{ from } \dots\dots\dots(4)$$

$$\text{from (1)} \Rightarrow P(1) = f(1) + g(1) = 0$$

129. The sum of 162th power of the roots of the equation $x^3 - 2x^2 + 2x - 1 = 0$ is :

समीकरण $x^3 - 2x^2 + 2x - 1 = 0$ के मूलों की 162^{वीं} घातों का योग होगा :

Ans. 3

Sol. The roots of the given equation

$$x^3 - 2x^2 + 2x - 1 = 0$$

$$\Rightarrow (x-1)(x^2 - x + 1) = 0$$

are 1, $-\omega$ and $-\omega^2$,

$$1^{162} + (-\omega)^{162} + (-\omega^2)^{162} = 3.$$

130. For $\alpha, \beta, z \in \mathbb{C}$ and $\lambda > 1$, If $\sqrt{\lambda-1}$ is the radius of the circle $|z-\alpha|^2 + |z-\beta|^2 = 2\lambda$, then $|\alpha-\beta|$ is equal to _____.

$\alpha, \beta, z \in \mathbb{C}$ तथा $\lambda > 1$ के लिए, यदि वृत्त $|z-\alpha|^2 + |z-\beta|^2 = 2\lambda$ की त्रिज्या $\sqrt{\lambda-1}$ है, तो $|\alpha-\beta|$ बराबर है _____

Ans. 2

Sol. $|z-z_1|^2 + |z-z_2|^2 = |z_1-z_2|^2$

$$z_1 = \alpha, z_2 = \beta$$

$$|\alpha-\beta|^2 = 2\lambda$$

$$|\alpha-\beta| = \sqrt{2\lambda}$$

$$2r = \sqrt{2\lambda}$$

$$2\sqrt{\lambda-1} = \sqrt{2\lambda}$$

$$\Rightarrow 4(\lambda-1) = 2\lambda$$

$$\lambda = 2$$

$$|\alpha-\beta| = 2$$

131. Let the positive numbers a_1, a_2, a_3, a_4 and a_5 be in a G.P. Let their mean and variance be $\frac{31}{10}$ and $\frac{m}{n}$ respectively, where m and n are co-prime. If the mean of their reciprocals is $\frac{31}{40}$ and $a_3 + a_4 + a_5 = 14$, then $m + n$ is equal to _____.

माना धनात्मक संख्याएँ a_1, a_2, a_3, a_4 तथा a_5 एक G.P. में हैं। माना इनके माध्य तथा प्रसरण क्रमशः $\frac{31}{10}$ तथा $\frac{m}{n}$ हैं, जहाँ m तथा n

सहअभाज्य हैं। यदि इन संख्याओं के व्युत्क्रमों का माध्य $\frac{31}{40}$ है तथा $a_3 + a_4 + a_5 = 14$ है, तो $m + n$ बराबर है _____

Ans. 211

Sol. Let $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$

$$\text{Given } \frac{a}{r^2} + \frac{a}{r} + a + ar + ar^2 = 5 \times \frac{31}{10} \quad \text{_____ (1)}$$

$$\text{and } \frac{r^2}{a} + \frac{r}{a} + \frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} = 5 \times \frac{31}{40} \quad \text{_____ (2)}$$

$$(1) \div (2) a^2 = 4 \Rightarrow a = 2$$

$$\therefore r + \frac{1}{r} = \frac{5}{2} (a \neq -2)$$

$$\Rightarrow r = 2$$

$$\therefore \text{Now } \frac{1}{2}, 1, 2, 4, 8$$

$$\therefore \sigma^2 = \frac{\sum x^2}{N} - \left(\frac{\sum x}{N} \right)^2$$

$$= \frac{186}{25} = \frac{M}{N} \Rightarrow 211 = m + n$$

132. If $\sum_{r=1}^{30} \frac{r^2 \binom{30}{r}}{\binom{30}{r-1}} = \alpha \times 2^{29}$, then α is equal to _____.

यदि $\sum_{r=1}^{30} \frac{r^2 \binom{30}{r}^2}{\binom{30}{r-1}} = \alpha \times 2^{29}$ है, तो α बराबर है _____।

Ans. 465

Sol. $\sum_{r=1}^{30} \frac{r^2 \binom{30}{r}^2}{\binom{30}{r-1}} = \alpha \times 2^{29}$

$$\sum r \left(\frac{\binom{30}{r}}{\binom{30}{r-1}} \right) \times r \cdot \binom{30}{r}$$

$$\sum r \times \left(\frac{30-r+1}{r} \right) \times r \times \frac{30}{r} \times {}^{29}C_{r-1}$$

$$= 30 \left(\sum_{r=1}^{30} (31-r) {}^{29}C_{r-1} \right)$$

$$= 30 \left(\sum_{r=1}^{30} 31 \times {}^{29}C_{r-1} - \sum_{r=1}^{30} r \cdot {}^{29}C_{r-1} \right)$$

$$= 30 \left(31 \sum_{r=1}^{30} {}^{29}C_{r-1} - \sum_{r=1}^{30} (r-1) {}^{29}C_{r-1} - \sum_{r=1}^{30} {}^{29}C_{r-1} \right)$$

$$= 30 (31 \times 2^{29} - 29 \times 2^{28} - 2^{29})$$

$$= 30 \times 31 \times 2^{28}$$

$$= 15 \times 31 \times 2^{29}$$

$$= 465 \times 2^{29}$$

133. Three urns A, B and C contain 4 red, 6 black; 5 red, 5 black; and λ red, 4 black balls respectively. One of the urns is selected at random and a ball is drawn. If the ball drawn is red and the probability that it is drawn from urn C is 0.4 then the square of the length of the side of the largest equilateral triangle, inscribed in the parabola $y^2 = \lambda x$ with one vertex at the vertex of the parabola, is _____.

तीन कलश A, B एवं C में क्रमशः 4 लाल, 6 काली; 5 लाल, 5 काली एवं λ लाल, 4 काली गेंद हैं। एक कलश यादृच्छया चुना जाता है तथा इसमें से एक गेंद निकाली जाती है। यदि निकाली गई गेंद लाल है तथा इसके कलश C से निकाले जाने की प्रायिकता 0.4 है, तो परवलय $y^2 = \lambda x$ के अंतर्गत सबसे बड़े समबाहु त्रिभुज, जिसका एक शीर्ष परवलय की शीर्ष पर है, की भुजा की लंबाई का वर्ग है।

Ans. 432

Sol. Urn A Urn B Urn C

Red	Black	Red	Black	Red	Black
4	6	5	5	λ	4

$$P\left(\frac{C}{R}\right) = \frac{P(C)P\left(\frac{R}{C}\right)}{P(A)P\left(\frac{R}{A}\right) + P(B)P\left(\frac{R}{B}\right) + P(C)P\left(\frac{R}{C}\right)}$$

$$0.4 = \frac{\frac{1}{3} \times \frac{\lambda}{(\lambda+4)}}{\frac{1}{3} \times \frac{4}{10} + \frac{1}{3} \times \frac{5}{10} + \frac{1}{3} \times \frac{\lambda}{(\lambda+4)}}$$

$$\Rightarrow \lambda = 6$$

$$\tan 30^\circ = 3t = \frac{3}{2}t^2$$

$$\frac{1}{\sqrt{3}} = \frac{2}{t}$$

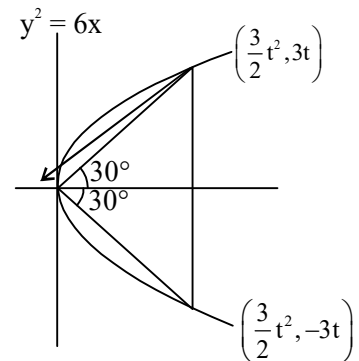
$$t = 2\sqrt{3}$$

$$\left(\frac{3}{2}t^2, 3t\right) = (18, 6\sqrt{3})$$

$$\ell^2 = 18^2 + (6\sqrt{3})^2$$

$$= 324 + 108$$

$$= 432$$



134. Let $w = z\bar{z} + k_1z + k_2iz + \lambda(1+i)$, $k_1, k_2 \in \mathbb{R}$. Let $\text{Re}(w) = 0$ be the circle C of radius 1 in the first quadrant touching the line $y = 1$ and the y -axis. If the curve $\text{Im}(w) = 0$ intersects C at A and B , then $30(AB)^2$ is equal to _____.

माना $w = z\bar{z} + k_1z + k_2iz + \lambda(1+i)$, $k_1, k_2 \in \mathbb{R}$ है। माना $\text{Re}(w) = 0$ प्रथम चतुर्थांश में इकाई त्रिज्या का एक वृत्त C है, जो रेखा $y = 1$ तथा y -अक्ष को स्पर्श करता है। यदि वक्र $\text{Im}(w) = 0$, वृत्त C को A तथा B पर काटता है, तो $30(AB)^2$ बराबर है

Ans. 24

Sol. $\omega = z\bar{z} + k_1z + k_2iz + \lambda(1+i)$

$$\text{Re}(\omega) = x^2 + y^2 + k_1x - x_2y + \lambda = 0$$

$$\text{Centre} \equiv \left(\frac{-k_1}{2}, \frac{k_2}{2} \right) \equiv (1, 2)$$

$$\Rightarrow k_1 = -2, k_2 = 4$$

$$\text{radius} = 1 \Rightarrow \lambda - 4 = 0$$

$$\text{Im} = k_1 y + k_2 x + \lambda = 0$$

$$\therefore 2x - y + 2 = 0$$

135. 25% of the population are smokers. A smoker has 27 times more chances to develop lung cancer than a non smoker. A person is diagnosed with lung cancer and the probability that this person is a smoker is $\frac{k}{10}$. Then the value of k is _____.

किसी जनसंख्या का 25% धूम्रपान करते हैं। एक धूम्रपान करने वाले को फेफड़े में कैंसर होने की संभावना धूम्रपान न करने वाले की अपेक्षा 27 गुना है। एक व्यक्ति के फेफड़े में कैंसर होने का पता चलता है तथा इस व्यक्ति के धूम्रपान करने की प्रायिकता $\frac{k}{10}$ है। तो k का मान है।

Ans. 9

Sol. $P(S) = \frac{1}{4}, P(N) = \frac{3}{4}$

$$P\left(\frac{C}{S}\right) = 27P\left(\frac{C}{N}\right); P\left(\frac{S}{C}\right) = \frac{P(S) \cdot P\left(\frac{C}{S}\right)}{P(S) \cdot P\left(\frac{C}{S}\right) + P(N) \cdot P\left(\frac{C}{N}\right)}$$

$$= \frac{\frac{1}{4} \times 27 \times P\left(\frac{C}{N}\right)}{\frac{1}{4} \times 27 \times P\left(\frac{C}{N}\right) + \frac{3}{4} \times P\left(\frac{C}{N}\right)} = \frac{27}{27+3} = \frac{27}{30} = \frac{9}{10} = \frac{k}{10} \Rightarrow k = 9$$

136. The product of the last two digits of $(1919)^{1919}$ is _____.

$(1919)^{1919}$ के अंतिम दो अंकों का गुणनफल है _____.

Ans. 63

Sol. $(1920-1)^{1919}$

$$= {}^{1919}C_0 (1920)^{1919} - \dots + {}^{1919}C_{1918} (1920) - {}^{1919}C_{1919}$$

$$= 100 [\lambda] + 1919 \times 1920 - 1$$

$$= 100\lambda + 3684880 - 1$$

$$100\lambda + 3684879$$

$$\text{Product of last two digits is } = 7 \times 9 = 63$$

137. Let A be the event that the absolute difference between two randomly chosen real numbers in the sample space $[0, 60]$ is less than or equal to a. If $P(A) = \frac{11}{36}$, then a is equal to _____.

माना प्रतिदर्श समष्टि $[0, 60]$ से यादृच्छया चुनी गई दो वास्तविक संख्याओं का निरपेक्ष अंतर a से कम या इसके बराबर होने की घटना

A है। यदि $P(A) = \frac{11}{36}$ है, तो a बराबर है।

Ans. 10

Sol. $|x - y| < a \Rightarrow -a < x - y < a$
 $\Rightarrow x - y < a$ and $x - y > -a$

$$P(A) = \frac{\text{ar}(\text{OACDEG})}{\text{ar}(\text{OBDF})}$$

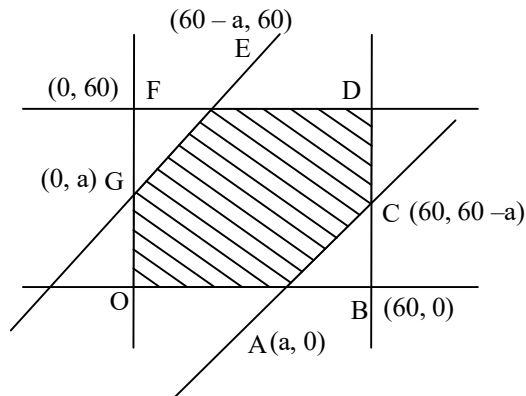
$$= \frac{\text{ar}(\text{OBDF}) - \text{ar}(\text{ABC}) - \text{ar}(\text{EFG})}{\text{ar}(\text{OBDF})}$$

$$\Rightarrow \frac{11}{36} = \frac{(60)^2 - \frac{1}{2}(60-a)^2 - \frac{1}{2}(60-a)^2}{3600}$$

$$\Rightarrow 1100 = 3600 - (60-a)^2$$

$$\Rightarrow (60-a)^2 = 2500 \Rightarrow 60-a = 50$$

$$\Rightarrow a = 10$$



138. The total number of numbers, lying between 100 and 1000 that can be formed with the digits 1, 2, 3, 4, 5, if the repetition of digits is not allowed and numbers are divisible by either 3 or 5, is _____.

अंकों 1, 2, 3, 4, 5 से 100 तथा 1000 के बीच की बनाई जा सकने वाली संख्याओं, यदि कोई भी अंक दोहराया नहीं जाता है तथा संख्याएँ या तो 3 से या 5 से विभाज्य हैं, की कुल संख्या है :

Ans. 32

Sol. Numbers divisible by 3

Case-I $1, 2, 3 \quad \underline{3} = 6$

Case-II $1, 3, 5 \quad \underline{3} = 6$

Case-III $2, 3, 4 \quad \underline{3} = 6$

Case-IV $3, 4, 5 \quad \underline{3} = 6$

Numbers divisible by 5

$$\boxed{} \boxed{} \boxed{5}$$

$$4 \times 3 \times 1 = 12$$

Common : 1, 3, 5 3, 1, 5

3, 4, 5 4, 3, 5

$$\text{Ans} = 24 + 12 - 4 = 32$$

139. If $\sum_{r=1}^{10} r!(r^3 + 6r^2 + 2r + 5) = \alpha(11!)$, then the value of α is equal to _____.

यदि $\sum_{r=1}^{10} r!(r^3 + 6r^2 + 2r + 5) = \alpha(11!)$ है, तो α का मान बराबर है _____ ।

Ans. 160

Sol. $\sum_{r=1}^{10} [r(r^3 + 6r^2 + 2r + 5)] = \alpha[11]$

$$\sum_{r=1}^{10} [r((r+1)(r+2)(r+3) - 9(r+1) + 8)]$$

$$\sum_{r=1}^{10} [r+3 - 9[r+1] + 8[r]$$

$$\sum_{r=1}^{10} [r+3 - [r+1] - 8[r+1] + 8[r]$$

$$\sum_{r=1}^{10} [r+3 - [r+1] - 8 \sum_{r=1}^{10} [r+1] - [r]$$

$$\Rightarrow [11(12 + 156 - 8)] = 160[11] = \alpha[11]$$

$$\alpha = 160$$

140. The mean of the coefficients of $x, x^2, \dots, x^7$ in the binomial expansion of $(2 + x)^9$ is _____.

$(2 + x)^9$ के द्विपद प्रसार में $x, x^2, \dots, x^7$ के गुणांकों का माध्य है।

Ans. 2736

Sol. Coefficient of $x = {}^9C_1 2^8$

$$\text{Of } x^2 = {}^9C_2 2^7$$

$$\text{Of } x^7 = {}^9C_7 \cdot 2^2$$

$$\text{Mean} = \frac{{}^9C_1 \cdot 2^8 + {}^9C_2 \cdot 2^7 + \dots + {}^9C_7 \cdot 2^2}{7}$$

$$= \frac{(1+2)^9 - {}^9C_0 \cdot 2^9 - {}^9C_8 \cdot 2^1 - {}^9C_9}{7}$$

$$= \frac{3^9 - 2^9 - 18 - 1}{7}$$

$$= \frac{19152}{7} = 2736$$